

MODULES 3A • 4A • 7A

Midterm Master Test Review

41 Review Questions

STUDENT EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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REVIEW

Test Two Master Test Review

41 questions worth focusing on as you prepare for Test 2, organized by subject. Each one tells you where to find the answer. Work through them in your own words, typing your answers on the lines.

Scientific Measurement

1. What are the three parts of a scientific measurement?

Find answers in: 3AE1

2. What is the confidence interval for 12.34 ± 0.02 cm? In percent, what probability is associated with the confidence interval of a single scientific measurement?

Find answers in: 3AE1

3. What is the difference between the least count and the tolerance of a measuring device? Do they always have the same value? Give an example.

Find answers in: 3AE1

4. How is a milliliter related to a cubic centimeter?

Find answers in: 3AE1

5. How many significant figures are in each of these measurements, written in standard notation: 0.081704 cm and 0.0069 cm?

Find answers in: 3AE2

6. The following is a scientific measurement written in scientific notation: 4.0560×10^3 cm. How many significant figures does this measurement have?

Find answers in: 3AE2

7. Multiply the following two measured quantities together, and state your answer to the correct number of significant figures, rounding up as necessary: 12.846×21.37 . What is your answer?

Find answers in: 3AE2

8. What is accuracy, and how is it different from precision?

Find answers in: 3AE2

9. In a scientific measurement, is being precise necessary, sufficient, or both, in order to be accurate? Explain.

Find answers in: 3AE2

- 10.** A nontrivial systematic error is an inherent limitation in what two aspects of an experiment? Give an example.

Find answers in: 3AE2

Astronomical Distances

- 11.** List astronomy's three distance units from smallest to largest. In miles, how big is an astronomical unit (au)?

Find answers in: 3AE3

- 12.** When is Earth closest to the Sun? If not distance, what causes the seasons?

Find answers in: 3AE3

- 13.** If light takes about 500 seconds to travel from the Sun to Earth, how many light-seconds is one au?

Find answers in: 3AE3

- 14.** Who discovered that the period of a Cepheid variable star (bright to dim to bright) could be used to find its distance? How are its period and luminosity related?

Find answers in: 3AE3

- 15.** What is the Chandrasekhar Limit, and why are Type Ia supernovae useful for measuring distance?

Find answers in: 3AE3

Gravity and Kepler's Laws

- 16.** What is a freely falling object?

Find answers in: 4AE1

- 17.** How are weight and mass different, and how are they related?

Find answers in: 4AE1

- 18.** Something weighs 150 pounds on Earth. How much does it weigh on the Moon, and does its mass change?

Find answers in: 4AE1

- 19.** According to Newton's Law of Universal Gravitation, the gravitational force between two masses is proportional to what?

Find answers in: 4AE1

- 20.** If the distance between the centers of two masses doubles, what happens to the gravitational force between them?

Find answers in: 4AE1

- 21.** State Newton's Third Law of Motion.

Find answers in: 4AE1

- 22.** At which lunar phases do spring tides and neap tides occur?

Find answers in: 4AE1

- 23.** Is Earth's measured acceleration due to gravity larger at the Equator or at the North Pole? State the approximate values.

Find answers in: 4AE1

- 24.** What did Kepler find about the shape of planetary orbits?

Find answers in: Kepler exercise

- 25.** What does Kepler's Second Law say, and what does it mean for a planet's speed?

Find answers in: Kepler exercise

- 26.** List the planets in order from the Sun outward.

Find answers in: 4AE1

- 27.** What is Pluto?

Find answers in: 4AE1

Our Cosmic Neighborhood

- 28.** Why do astronomers use the 24-hour clock and UTC? How far behind UTC is a city at 90° W?

Find answers in: 7AE1

- 29.** What does equinox mean?

Find answers in: 7AE1

- 30.** Where does the Sun rise and set on an equinox, and how is the setting Sun's path related to an observer's latitude? What is Chicago's latitude?

Find answers in: 7AE1

- 31.** Which three stars form the Winter Triangle?

Find answers in: 7AE1

- 32.** How do you find Orion, and which star is his left shoulder as he faces away from you?

Find answers in: 7AE1

- 33.** How many moons does Mars have, and what are they called? What gives Mars its red color?

Find answers in: 7AE2

- 34.** What is the tallest volcano in the Solar System, and where is it?

Find answers in: 7AE2

- 35.** Why is Venus seen only as a morning or evening star? Give its ancient names.

Find answers in: 7AE2

- 36.** Why does the surface of Venus stay near 867 °F, both day and night?

Find answers in: 7AE2

- 37.** How did the Moon form, according to the giant-impact model? How far away is it now?

Find answers in: 7AE2

- 38.** What does it mean that the Moon is tidally locked, and is its far side always dark?

Find answers in: 7AE2

- 39.** Distinguish a sidereal month from a lunation.

Find answers in: 7AE2

- 40.** List the Moon's phases in order, starting from New Moon. Which phase comes between New Moon and First Quarter?

Find answers in: 7AE2

- 41.** What kind of galaxy is the Milky Way? Name the five largest members of the Local Group.

Find answers in: 7AE2
